Homework 3

Kierra Miller

2026-05-27

Table of contents

Part 1. Set Up

```
# Homework Set Up and install packages
library(tinytex)
library(tidyverse)
library(here)
library(janitor)
library(readxl)

# creating object and reading in personal data set

my_data <- read_csv("datasheet.csv",
# editing format of column names
show_col_types = FALSE,
                    col_names = TRUE)

# clean names
my_data_clean <- my_data %>%
  clean_names()
# Make sure ounces_consumed is numeric
my_data_clean <- my_data_clean %>%
  mutate(
    ounces_consumed = as.numeric(ounces_consumed)
  )
# Standardize class location names
my_data_clean <- my_data_clean %>%
  mutate(
    class_location = recode(
      class_location,
      "Givetz" = "Girvetz"
    )
  )
# insert kelp dataset
kelp <- read_csv(here("data", "temp-kelp.csv"))
```

Part 2. Problems

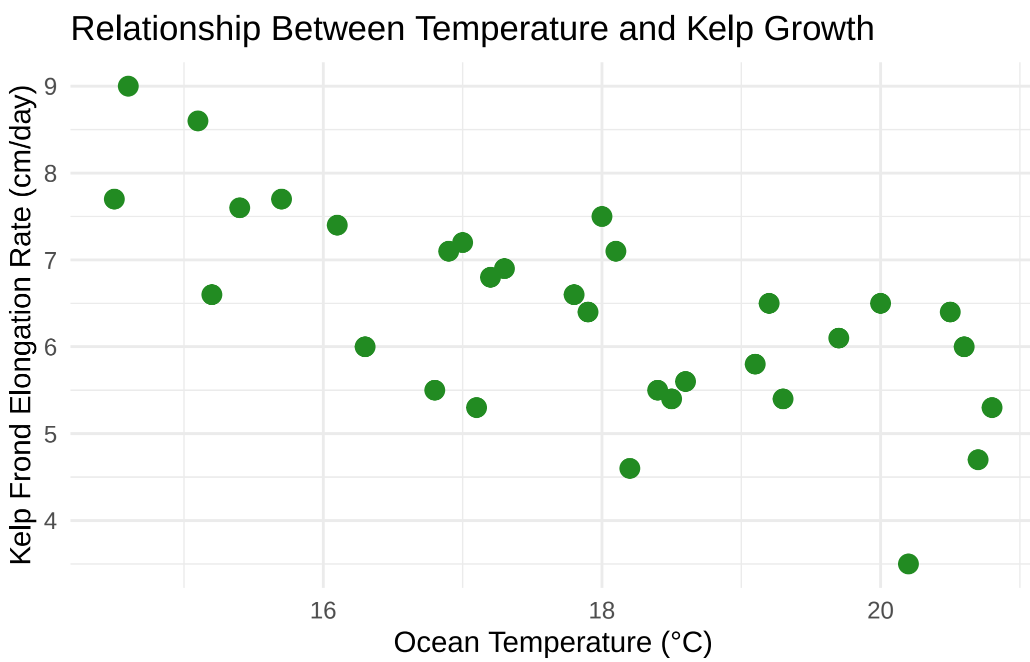
Problem 1. Giant Kelp Fronds

a. Appropriate test

The most appropriate test is a Pearson correlation because both temperature and kelp elongation rate are continuous variables. Pearson correlation measures the strength and direction of the linear relationship between two variables. Because the goal is to determine whether temperature is associated with kelp growth rather than predict kelp growth from temperature, correlation is more appropriate than linear regression.

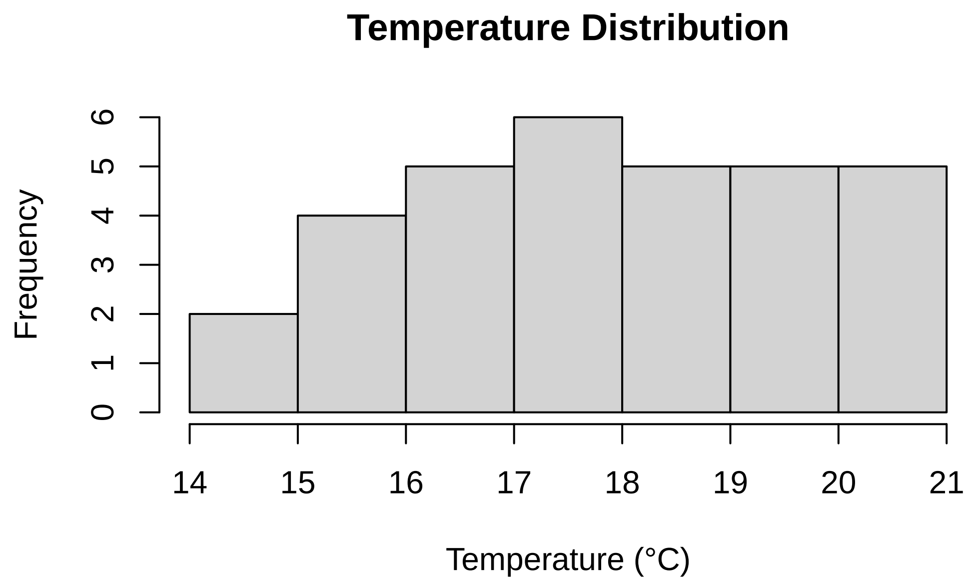
b. Visualization

```r
# create scatterplot
ggplot(kelp, aes(x = temp_c, y = kelp_elong)) +
# plot observations
  geom_point(color = "forestgreen", size = 3) +
  # insert labels for axes and title
  labs(
    x = "Ocean Temperature (°C)",
    y = "Kelp Frond Elongation Rate (cm/day)",
    title = "Relationship Between Temperature and Kelp Growth"
  ) +
  # insert theme
  theme_minimal()
```

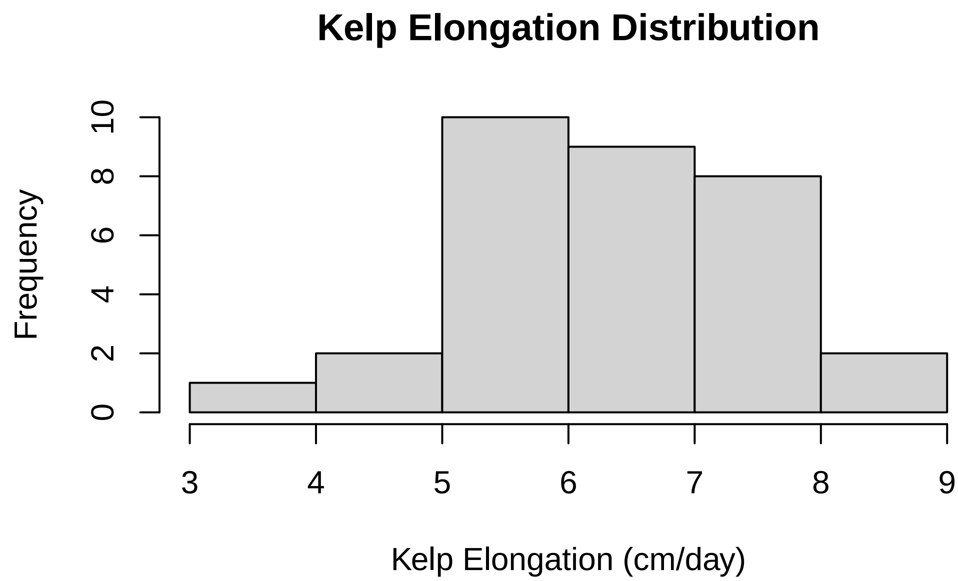


c. Check your assumptions and run test

```
# check tempertature distribution
hist(kelp$temp_c,
     main = "Temperature Distribution",
     xlab = "Temperature (°C)")
```




```r
# check kelp distribution
hist(kelp$kelp_elong,
     main = "Kelp Elongation Distribution",
     xlab = "Kelp Elongation (cm/day)")
```



A scatterplot was used to assess whether the relationship between temperature and kelp elongation rate was approximately linear. Histograms were used to examine the distributions of both temperature and kelp elongation rate. The scatterplot suggests a roughly linear relationship, and both variables appear reasonably normally distributed with no extreme outliers. Therefore, the assumptions for Pearson correlation appear to be met.

```
# Run correlation test
cor.test(kelp$temp_c, kelp$kelp_elong)
```

```

	Pearson's product-moment correlation

data:  kelp$temp_c and kelp$kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
       cor 
-0.6877489 
```

d. Results communication

A Pearson correlation showed a significant negative relationship between temperature and kelp elongation rate (Pearson's $r = -0.69$, $t(30) = -5.19$, $p < 0.001$, $= 0.05$). This indicates that higher temperatures are associated with lower kelp growth rates. The correlation coefficient suggests a moderate to strong negative relationship between the two variables.

e. Test implications

These results suggest that increasing ocean temperatures may impact kelp growth rates. Since the relationship appears to be negative, warming oceans could reduce kelp productivity, affecting ecosystem health and function. This information can help guide conservation and climate adaptation and management strategies.

f. Double check work

```r
# spearman correlation test to double check work
cor.test(
  kelp$temp_c,
  kelp$kelp_elong,
  method = "spearman"
)
```

```
	Spearman's rank correlation rho

data:  kelp$temp_c and kelp$kelp_elong
S = 9216.1, p-value = 1.29e-05
alternative hypothesis: true rho is not equal to 0
sample estimates:
       rho 
-0.6891684 
```

A Spearman rank correlation was used as a secondary analysis. The Spearman correlation produced the same overall conclusion as the Pearson correlation, indicating a negative relationship between temperature and kelp elongation rate. This agreement between methods increases confidence in the result.

Problem 2. Personal Data

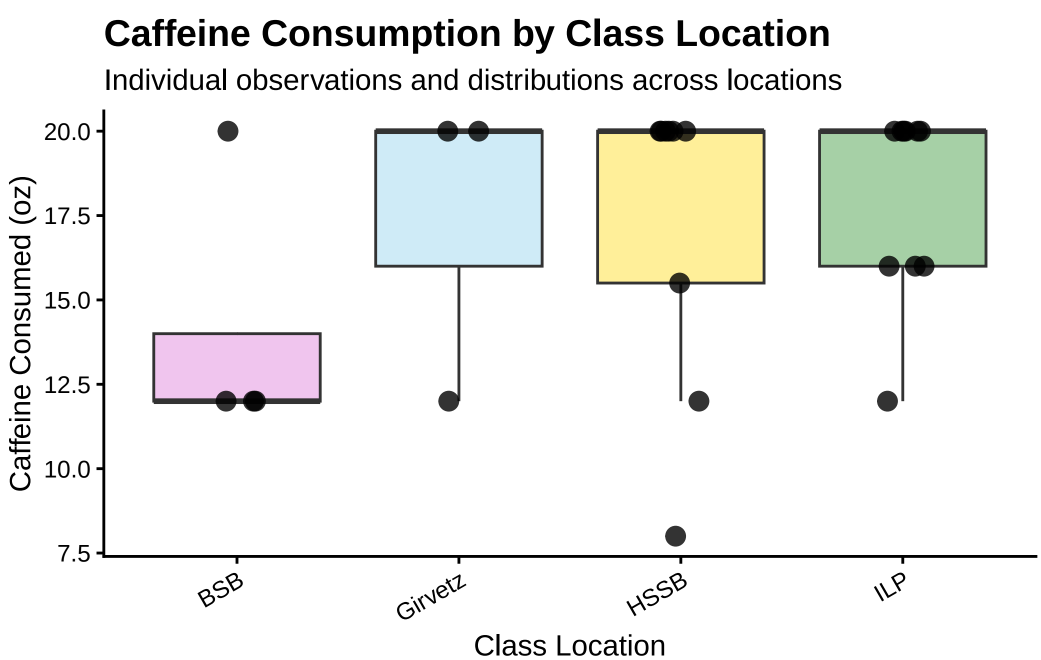
a. Update plots

```r
# Updated visualization plots from Homework 2
# Figure 1: Caffeine consumption by class location attendance
# create boxplot and jitter plot of caffeine consumption by class location
ggplot(
  my_data_clean,
  aes(
    x = class_location,
    y = ounces_consumed,
    fill = class_location
  )
) +

  # Show distribution of observations within each location
  geom_boxplot(
    alpha = 0.4,
    outlier.shape = NA
  ) +

  # Show individual observations
  geom_jitter(
    width = 0.10,
    height = 0,
    size = 3,
    alpha = 0.8
  ) +
# assign custom colors
  scale_fill_manual(
  values = c(
    "BSB" = "orchid",
    "Girvetz" = "skyblue",
    "HSSB" = "gold",
    "ILP" = "forestgreen"
    )
  ) +
# add labels for title, subtitle, axes
  labs(
    title = "Caffeine Consumption by Class Location",
```

```
    subtitle = "Individual observations and distributions across locations",
    x = "Class Location",
    y = "Caffeine Consumed (oz)"
  ) +
# apply theme
  theme_classic() +
# customize appearance
  theme(
    legend.position = "none",
    plot.title = element_text(
      size = 14,
      face = "bold"
    ),
    axis.text.x = element_text(
      angle = 30,
      hjust = 1
    )
  )
```




```
# updating Plot 2 (Continuous)

# Plot 2: Caffeine Consumption by Class Type
```

```r
# create plot showing caffeine consumption by class subject
ggplot(
  my_data_clean,
  aes(
    x = type_of_class,
    y = ounces_consumed,
    fill = type_of_class
  )
) +

  # show individual observations
 geom_jitter(
  width = 0.1,
  height = 0,
  size = 3,
  alpha = 0.8
) +
 # create labels for title, subtitle, axes
  labs(
    title = "Caffeine Consumption by Class Type",
    subtitle = "Daily caffeine intake across different types of classes",
    x = "Class Subject",
    y = "Caffeine Consumed (oz)"
  ) +
# apply classic theme
  theme_classic() +
# customize appearance
  theme(
    legend.position = "none",
    plot.title = element_text(
      size = 14,
      face = "bold"
    ),
    axis.text.x = element_text(
      angle = 30,
      hjust = 1
    )
  )
```

b. Captions

Figure 1. Caffeine consumption (oz) across class locations. Each point represents one observation day, and horizontal jitter was applied to reduce overlap while preserving the observed caffeine consumption values. The boxplots summarize the distribution of caffeine consumption at each location. Differences among locations may indicate variation in caffeine intake across learning environments.

Figure 2. Caffeine consumption (oz) across class subjects. Each point represents one observation day, and horizontal jitter was applied to reduce overlap while preserving the observed caffeine consumption values. Some class subjects appear to be associated with higher caffeine consumption, although additional observations would be needed to determine whether these patterns are consistent.

Problem 3. Affective visualization

a. Describe in words what an affective visualization could look like for your personal data

An affective visualization of my data would highlight how caffeine consumption changes based on academic workload and environment. Instead of focusing only on numerical values, the visualization would emphasize how different days feel, using color intensity or size to represent higher or lower caffeine intake. For example, days with more assignments due could be represented with darker colors to convey increased stress or fatigue, and the size of the point represented by a coffee cup could be larger to represent days with a higher number of classes. This approach would help communicate not just the data, but the experience behind the collection process.

b. Create a sketch (on paper) of your idea

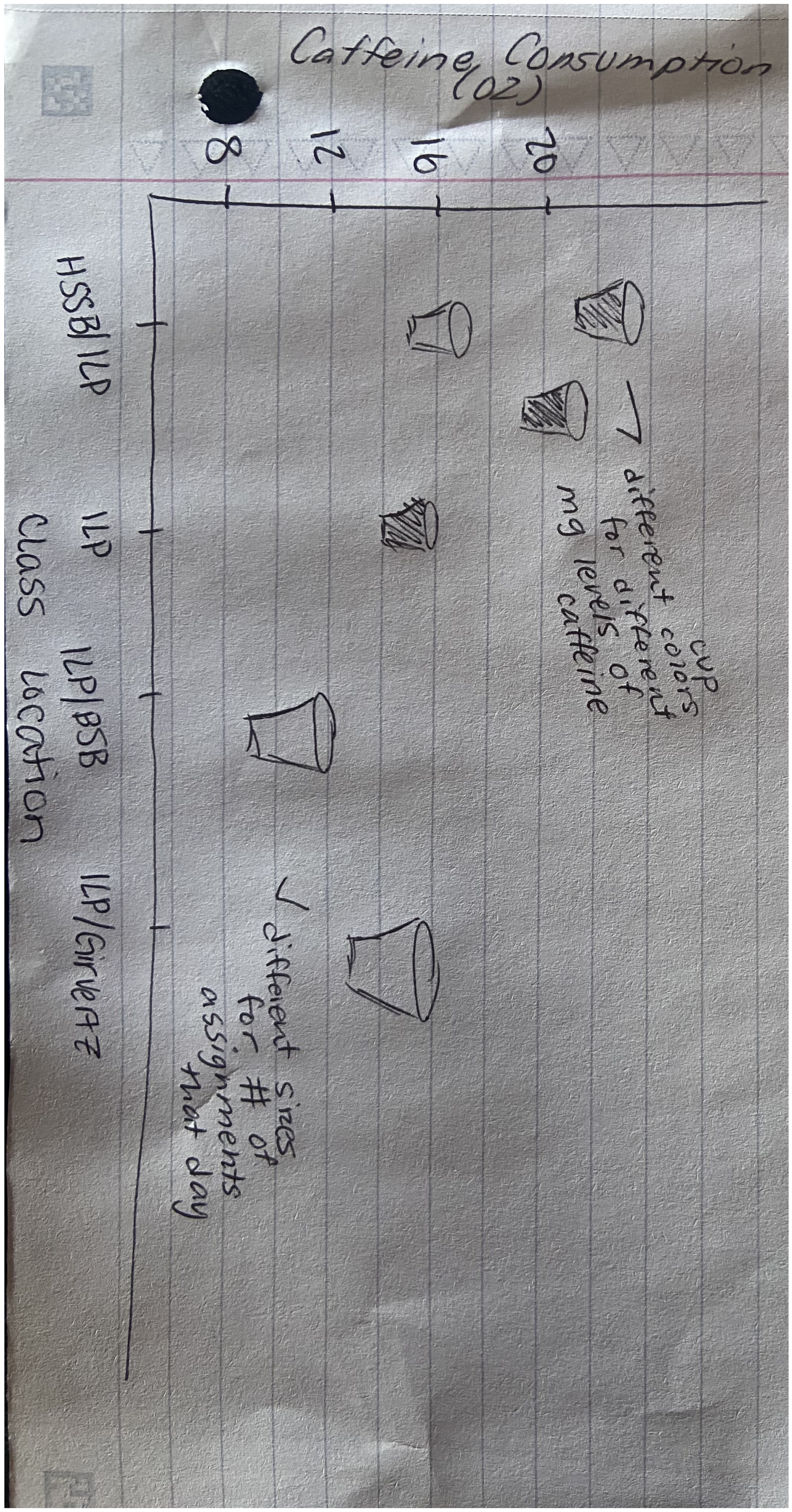


Figure 1: Sketch of personal data visualizaton

c. Make a draft of the visualization

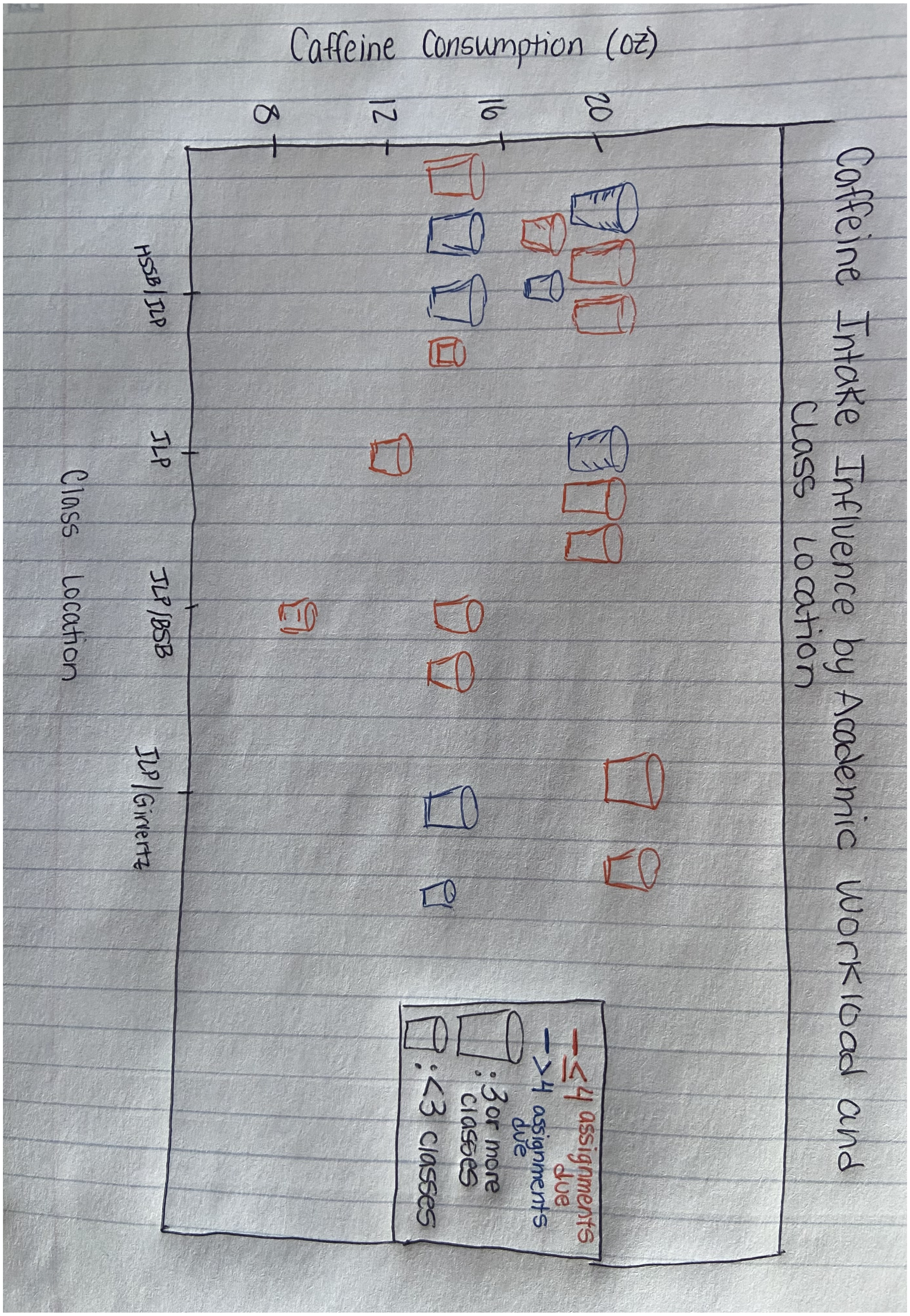


Figure 2: Sketch of personal data visualizaton

d. Write an artist statement

Content

This visualization represents daily caffeine consumption in relation to academic workload and class environment, highlighting how different days influence energy needs.

Influences

I was inspired by projects such as the Dear Data reference project, which incorporated personal story telling into data and statistics, creating a unique visual expression.

Form

The piece is a hand-drawn visualization that uses color and sizing to represent variation in caffeine intake based on class environment and academic workload. I used critiques from other classmates to help improve my final product. I will be using R to create the base visualization resembling a typical graph, and then I will be adding artistic elements over a printed copy of the graph showcasing how art and statistics can work together to fully show the data and results to an audience.

Process

I began by thinking of how I wanted to display the data I have collected, then I decided to begin sketching how to represent my data visually, I choose to use shapes and colors to translate my dataset into a more expressive format using drawing and blending a traditional figure with creative features.

e. Prep materials to share with class

Presentation slides

View my slides

Problem 4. Statistical critique

a. Revisit and summarize

The authors investigated how environmental variables along temperate rocky coastlines influence kelp forest distribution and biomass. Their main hypothesis was that kelp presence and biomass could be predicted from environmental conditions such as sea surface temperature, nutrient availability, depth, and wave exposure. To address this question, the authors used Analysis of Variance (ANOVA) tests to compare kelp biomass across different environmental conditions and geographic regions. ANOVA allowed the authors to determine whether mean kelp biomass differed significantly among environmental groups and identify which factors were most strongly associated with kelp distribution and productivity.

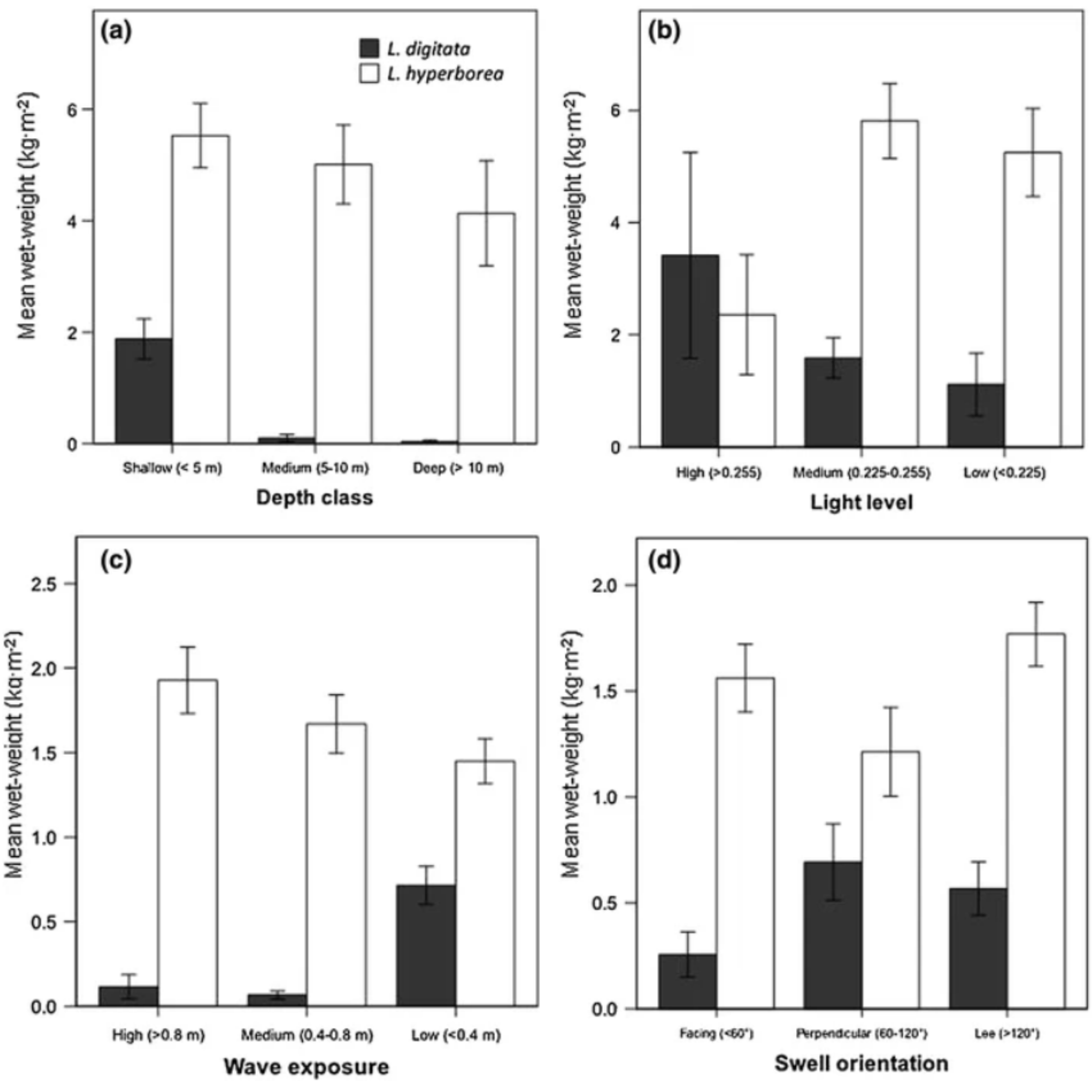


Figure 3: Figure showing kelp biomass differences among environmental groups

b. Visual clarity

The figure presents the ANOVA results clearly by displaying kelp biomass across environmental groups with labeled axes and an appealing layout. The figure effectively communicates differences among groups using summary statistics, making it easy to compare mean biomass values. However, the underlying raw observations are not shown, making it difficult to assess variability in the groups and the spread of the data. Including individual data points would provide additional context for interpreting the statistical results.

c. Aesthetic clarity

The figure has a relatively good data-to-ink ratio because most visual elements contribute directly to communicating differences in kelp biomass among environmental groups. However, the emphasis is placed primarily on group summaries rather than the underlying observations. Because only summary statistics are displayed, some information about the distribution of the data is lost. The figure remains readable, but showing the raw data would improve transparency without substantially increasing visual clutter.

d. Recommendations

I would recommend adding the individual kelp biomass observations to the figure using jittered points overlaid on the group summaries. This would allow readers to evaluate variation within each environmental group and better understand the strength of the patterns reported by the ANOVA. I would also use distinct but colorblind-friendly colors for each environmental group so that comparisons can be made more easily. These changes would improve interpretability while maintaining a relatively simple and uncluttered design.